RevInsight AI: AI-Driven Revenue Concentration Analysis System for Manoy's Motorcycle Parts, Accessories and Services

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In a Partial Fulfillment
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Bachelor of Science in Information Technology
with Specialization Track in Business Analytics

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This capstone project titled: "RevInsight AI: AI-Driven Revenue Concentration Analysis System for Manoy's Motorcycle Parts, Accessories and Services", prepared and submitted by Bunsay, Axel Stephan E., Deroxas, Eliza T., Robiso, Krenz Gerome E., in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with specialization track in Business Analytics, has been examined and is recommended for acceptance and approval for oral examination.

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1. INTRODUCTION

This chapter discussed the context of the problem, objectives, significance, and the scope and limitations.

1.1 Background of the Study

In many communities, motorcycles serve as one of the primary modes of transportation because they are affordable, efficient, and accessible. Consequently, the motorcycle parts, accessories, and services sector play a vital role in supporting daily transportation needs by providing essential products and services such as maintenance, repair, and performance enhancement for motorcycle users.

Currently, many small and medium-sized enterprises (SMEs), including motorcycle parts and service shops, still rely on manual or semi-digital methods such as handwritten records or basic spreadsheets to track sales transactions. While these methods allow businesses to record data, they often lack the capability to effectively organize, analyze, and interpret sales information. As a result, business owners experience difficulty in identifying which products or services contribute the most to their revenue and whether their income is heavily dependent on a limited number of sources. Studies confirm that SMEs adopting business intelligence and analytics achieve higher efficiency, improved customer service, and stronger revenue growth compared to those relying on traditional methods (Alsulami et al., 2024; Tsiu et al., 2025).

These limitations lead to several operational challenges. The absence of a structured analytical system prevents business owners from detecting high-performing and underperforming products, identifying sales trends, and evaluating the distribution of revenue streams, which results in largely intuitive rather than data-driven decision-making. Furthermore, without proper analysis of revenue concentration, businesses may unknowingly rely too heavily on specific products or services, making them vulnerable to financial instability if demand for those items' declines. Research on SME inventory management highlights that analytics-driven practices reduce risks such as cash flow problems and stockouts, thereby improving overall performance (Panigrahi et al., 2024).

Manoy's Motorcycle Parts, Accessories, and Services, established in 2021, operates under similar conditions. Although the business maintains records of its sales transactions, it does not currently have a system capable of analyzing revenue distribution or assessing the structural stability of its income sources. This limitation restricts the owner's ability to gain meaningful insights into business performance, identify growth opportunities, and mitigate risks related to revenue concentration.

With the advancement of technology, artificial intelligence (AI) and data analytics have emerged as powerful tools for improving business decision-making. AI-enabled systems can process large volumes of data, detect patterns, and generate meaningful insights that support strategic planning. Recent studies emphasize that AI adoption in SMEs not only enhances revenue growth but also strengthens competitive advantage when integrated with big data and IoT technologies (Ardito et al., 2024; Oldemeyer et al., 2024).

In response to these challenges, this study proposes the development of RevInsight AI: An AI-Driven Revenue Concentration Analysis System for Manoy's Motorcycle Parts, Accessories, and Services. The proposed system aims to collect, store, and organize sales data within a centralized platform while utilizing AI-driven analytics to evaluate revenue distribution and identify concentration risks. By transforming raw transactional data into actionable insights, the system seeks to improve decision-making, optimize business performance, and support long-term operational stability.

1.2 Statement of the Problem

This study aims to address the difficulty of Manoy's Motorcycle Parts, Accessories and Services in organizing, analyzing, and interpreting its sales and service revenue records. Specifically, it seeks to answer the following questions:

1. How can the current sales and service records of the business be organized in a centralized system?
2. How can the system identify the top revenue-generating motorcycle parts, accessories, and mechanic labor services?
3. How can the system determine whether revenue is concentrated in only a few products, accessories, or services?
4. How can OpenAI API-assisted business insight summaries help the owner understand revenue concentration patterns and possible business risks?
5. How acceptable is the developed system in terms of functionality, usability, performance, accuracy, and overall usefulness?

1.3 Objectives of the Study

The general objective of this study is to design and develop RevInsight AI, an AI-driven revenue concentration analysis system for Manoy's Motorcycle Parts, Accessories and Services that supports data-driven business decision-making through organized sales and service revenue analysis.

Specifically, this study aims to:

1. Develop a centralized system that manages business transaction data, including:

   1.1 Collection of sales records from motorcycle parts, accessories, and mechanic labor services;

   1.2 Storage and organization of transaction records in a centralized database; and

   1.3 Provision of accessible and structured records for analysis and reporting.

2. Implement revenue concentration analysis features, including:

   2.1 Identification of products, accessories, and labor services contributing the highest revenue;

   2.2 Computation of revenue distribution using percentage-share analysis to determine concentration levels; and

   2.3 Detection of revenue dependence on limited products, accessories, or services based on defined concentration indicators.

3. Integrate OpenAI API-assisted business insight summaries that explain computed revenue analytics, including top contributors, concentration patterns, and possible business risks for owner review.

4. Evaluate the developed system in terms of functionality, usability, performance, accuracy, and overall user acceptance.

1.4 Scope and Delimitations

The study covers the design and development of an AI-driven revenue concentration analysis system for Manoy's Motorcycle Parts, Accessories, and Services. It includes collection, storage, processing, and visualization of sales transactions and the use of AI-based analytics to identify revenue concentration, analyze trends, and generate insights through dashboards and reports. Access will be provided to the business owner and authorized personnel for monitoring and decision-making.

The research is limited to basic description and analysis of existing sales data. It does not include forecasting, complex machine learning model development, external financial forecasting, or integration with external systems such as supplier databases and third-party financial tools. System testing will involve a small group of users from the client, so findings may not fully represent performance across the broader industry.

The study will also observe data privacy and ethical handling of business transaction records. Data collected and stored by the system will be limited to transaction-related business information needed for revenue analysis, such as product and service details, transaction dates, amounts, revenue categories, and related operational records. Customer-identifying information will be minimized and will not be used for research outputs unless necessary and permitted by the client. Access to the centralized database will be restricted to the business owner and authorized personnel through login authentication and role-based access control. Any data used for research, testing, demonstration, or evaluation will be handled with client consent and protected from unauthorized access, disclosure, alteration, or misuse. The researchers will follow applicable data protection standards, including the principles of transparency, legitimate purpose, and proportionality under the Data Privacy Act of 2012 (National Privacy Commission, n.d.).

1.5 Significance of the Study

This study is significant as it introduces a system that helps a motorcycle parts and services business organize records, monitor revenue distribution, and interpret concentration risks through dashboards, reports, and OpenAI API-assisted summaries.

Business Owner. The system will help the owner identify whether the business depends heavily on a small number of motorcycle parts, accessories, or mechanic labor services. This can support better decisions on restocking, pricing, service promotion, mechanic workload monitoring, and risk management.

Employees and Mechanics. The system will support employees and mechanics by providing a structured way to record product sales, service transactions, labor fees, and job-related information. This can reduce manual checking, improve record retrieval, and make service-related revenue easier to monitor.

Customers. The system will provide customers with a more organized way to view available products or services, place orders or service requests, and monitor order or service status when applicable. This can improve the transaction process and reduce delays caused by manual coordination.

Business Sectors, particularly SMEs. The study may serve as an example of how small businesses can use practical analytics to understand revenue dependence without implementing complex machine learning models. It demonstrates how transaction records can be transformed into useful business information through dashboards and API-assisted summaries.

Researchers and Future Developers. The study may serve as a reference for future capstone projects involving sales analytics, service revenue monitoring, dashboard development, and responsible AI API integration in small business settings.

1.6 Definition of Terms

Artificial Intelligence (AI) - The use of an approved AI service, specifically the OpenAI API, to generate natural-language business insight summaries from computed revenue analytics.

Dashboard - A visual interface that presents key revenue metrics, charts, summaries, and reports for business monitoring and decision-making.

Data Analytics - The process of examining and interpreting transaction data to identify revenue patterns, top contributors, and concentration levels.

Decision Support System (DSS) - An information system that provides organized data, analytics, dashboards, and insight summaries to support human decision-making.

Revenue Concentration - The degree to which business income depends on a limited number of products, accessories, services, or labor categories.

Sales Data - Transaction records that include product or service details, quantities, prices, dates, and revenue amounts used as the basis for analytics.

System Architecture - The structured arrangement of system components, including the interface, backend, database, analytics module, AI API integration, and reporting module.

User Acceptance Testing (UAT) - An evaluation process where intended users test the system to determine whether it meets business needs and usability expectations.

Usability - The degree to which users can operate the system easily, efficiently, and satisfactorily.

Visualization - The presentation of data through charts, graphs, dashboards, and visual reports to make revenue information easier to understand.

2. LITERATURE REVIEW

This chapter presents the related literature, systems, technologies, and conceptual framework that support the development of RevInsight AI for revenue concentration analysis.

2.1 Related Systems

Gonçalves et al. (2023) developed integrated performance dashboard visualizations using Power BI as a business intelligence platform. The system presents business indicators through visual dashboards, interactive charts, and organized performance views that help users analyze sales-related information more quickly. Its main functionality is the conversion of raw business data into understandable dashboard outputs for monitoring and decision-making. However, the system is mainly designed as a general BI dashboard and does not focus on revenue concentration among specific products, accessories, and labor services in a small motorcycle-related business. It also depends on prepared business intelligence data rather than a transaction-focused system that directly records daily sales and service income. This limitation shows the need for RevInsight AI, which will combine transaction recording, revenue categorization, concentration analysis, and API-assisted business insight summaries in one client-specific system.

Nurdin et al. (2022) presented a business intelligence-based sales information system for an avocado distributor. The system uses sales data visualization, sales summaries, and forecasting features to identify product performance and support planning. Its features allow the business to determine which products generate strong sales and which items require attention. However, the system is designed for a distributor setting and focuses mainly on product sales, not on a mixed business operation involving parts, accessories, and mechanic labor services. It also does not emphasize revenue concentration risk, where a business may become dependent on only a few income sources. This gap supports the development of RevInsight AI because the proposed system will classify revenue sources, show top-performing items and services, and generate AI-supported summaries that explain concentration patterns in a simple and owner-friendly manner.

Wahyudi and Hendriawan (2024) designed a web-based sales transaction and reporting system for a motorcycle spare parts company. The system was developed to support transaction processing, stock management, and sales reporting, which are important functions for businesses that handle motorcycle-related products. Its features help automate sales records and generate reports that can reduce manual checking and improve operational efficiency. However, the system focuses mainly on transaction and report generation and does not emphasize revenue concentration analysis, service labor revenue, or AI-supported explanation of business trends. This gap supports the development of RevInsight AI because the proposed system will not only record product and service transactions but also analyze revenue distribution, identify top-performing products and labor services, and generate API-assisted business insight summaries for owner review.

Ogbuefi et al. (2024) discussed real-time data visualization and analytics systems for small and medium-sized enterprises. These systems transform fragmented operational data into dashboards that display key performance indicators, trends, and business patterns. They are useful for inventory control, marketing analysis, financial planning, and management reporting. However, the study also noted challenges such as inconsistent data quality, lack of technical skills, and difficulty in interpreting analytics results among SME users. These limitations are important for RevInsight AI because the client also needs a system that is not only analytical but also understandable to a business owner. The proposed system addresses this gap by organizing sales and labor records into clear dashboard outputs and using an approved AI API only to generate plain-language insight summaries from computed results, instead of requiring the client to interpret complex analytics independently.

Bintang and Jatmiko (2026) designed and developed an integrated ordering and sales analysis system for Kaluna Living, a micro-enterprise that previously relied on manual workflows for order recording, custom request processing, and monthly sales report preparation. The system supports centralized ordering, custom request handling with image upload, and sales insight visualization for monthly revenue and order volume. It was developed using the SDLC Prototyping model and evaluated with the System Usability Scale, producing an average score of 81.66, which indicates excellent usability. However, the system is focused on order management and sales summary visualization for a handcrafted product business, not on revenue concentration among motorcycle parts, accessories, and mechanic labor services. It also does not include AI/API-assisted natural-language explanations of revenue dependency. This gap supports RevInsight AI because the proposed system will organize product and service transactions, compute revenue concentration, identify top revenue contributors, and generate concise AI-supported business summaries for owner review.

Zamil (2025) reviewed AI-driven business analytics for financial forecasting and decision support models in SMEs. The review shows that artificial intelligence can support revenue prediction, budgeting, cash-flow estimation, and financial planning when enough reliable business data are available. These findings are useful because they show the value of AI-supported analytics in improving financial decision-making. However, many systems discussed in the review involve machine learning, deep learning, or complex predictive models, which are beyond the required scope of a BSIT capstone project. This limitation guides the scope of RevInsight AI: the proposed system will not develop or train a new machine learning model. Instead, it will use computed analytics and an approved AI API to produce natural-language business insight summaries based on revenue concentration results, top revenue sources, and dashboard data.

Okeke et al. (2024) examined financial stability in SMEs through strategic budgeting and revenue management. Their study emphasizes the importance of pricing, revenue stream diversification, and continuous monitoring of income sources to reduce financial risk. The system concepts discussed in the study are relevant because they show how SMEs can benefit from tools that identify revenue dependence and support better planning. However, the study is broader in scope and does not present a client-specific transaction system for daily sales, service labor monitoring, or concentration visualization. It also does not provide a practical workflow for small businesses that need simple reports instead of advanced financial systems. This gap supports RevInsight AI because the proposed system will help the owner identify whether revenue is concentrated in limited products or services and will summarize the results through API-assisted insights.

Elumilade et al. (2025) studied the use of financial data analytics for business growth, fraud prevention, and risk mitigation. Their work explains how financial analytics systems can help organizations identify patterns, support scenario analysis, and detect possible irregularities in financial data. The strength of these systems is their ability to convert large financial datasets into information that supports planning and risk management. However, the study is oriented toward broader financial markets and large-scale financial data, while RevInsight AI focuses on a smaller client with simpler operational needs. It does not need complex fraud detection or advanced predictive modeling. Instead, the gap is the need for a practical revenue analytics system that can record transactions, compute revenue shares, identify top contributors, and use API-assisted summaries to explain the concentration results to the business owner.

Drydakis (2022) analyzed the role of artificial intelligence in reducing SME business risks during the COVID-19 pandemic. The study found that AI applications used in pricing, sales, and cash-flow forecasting can help SMEs improve decision-making and reduce uncertainty. This is related to RevInsight AI because revenue concentration analysis also supports risk awareness by showing whether income depends heavily on a few items or services. However, the study focuses on AI as a broader business-risk tool and discusses forecasting and anomaly detection functions that may require more advanced datasets and algorithms. For the proposed system, the AI component will be limited to decision-support explanation. RevInsight AI will rely on system-computed totals, rankings, and concentration indicators, then use an approved AI API to generate understandable summaries, observations, and possible action points for human review.

Perdana et al. (2022) examined data analytics in small and mid-size enterprises and identified information quality and system quality as important factors that affect analytics value. Their study shows that analytics systems are more useful when the data are accurate, complete, accessible, and presented in a way that supports business decisions. However, SMEs may experience barriers such as limited understanding of analytics, lack of technical expertise, and concerns about data security. These findings are directly relevant to RevInsight AI because the proposed system must be simple, reliable, and useful for a local business owner. The gap addressed by RevInsight AI is the need for a centralized system that improves data quality through structured transaction recording, provides clear revenue dashboards, and uses API-assisted business summaries to make analytics easier to understand.

Campbell and Oyinloye (2024) discussed the use of real-time business intelligence and predictive analytics to support the financial stability of SMEs in economically distressed regions. Their work shows that dashboards and analytics can improve forecasting accuracy, reduce operating costs, support customer retention, and increase business performance. However, the study presents a broad analytics adoption blueprint rather than a specific system for revenue concentration analysis in a small retail and service business. Predictive analytics may also require technical capacity and historical data that may not be immediately available to the client. RevInsight AI responds to this gap by focusing first on practical descriptive analytics: transaction recording, category-based revenue computation, top-performing products and labor services, and AI-supported summaries generated from the system's computed results rather than from a newly developed predictive model.

Dachepalli (2025) discussed AI-driven decision support systems in enterprise resource planning environments. Such systems use artificial intelligence to help interpret business data, generate recommendations, and support managerial decisions. Their value lies in transforming enterprise data into useful decision-support outputs. However, ERP-based decision support systems are often large, expensive, and designed for organizations with complex departments and extensive data sources. This makes them less practical for a small business that mainly needs sales recording, service revenue monitoring, and simple revenue concentration analysis. The limitation supports the development of RevInsight AI as a lightweight and client-specific alternative. Rather than building an ERP or a machine learning model, RevInsight AI will compute revenue analytics internally and use an approved AI API only to generate concise, human-readable business insight summaries.

2.2 Related Technologies

Technical Background

The development of RevInsight AI focuses on creating an AI-supported revenue concentration analysis system that will help Manoy's Motorcycle Parts, Accessories and Services monitor, organize, and evaluate revenue performance across products and services. The system aims to determine how business revenue is distributed and to identify possible risks caused by overdependence on limited income sources.

The technical foundation of the project involves web-based development, database management, analytics processing, dashboard visualization, and AI-supported insight generation. The system uses frontend technologies for interface presentation, backend technologies for request processing, a database for centralized storage, visualization tools for dashboards, and an approved AI API for generating natural-language explanations from summarized revenue analytics.

Frontend Frameworks

Frontend frameworks are important in developing responsive, interactive, and user-friendly interfaces for modern web systems. RevInsight AI uses React.js and JavaScript to create transaction forms, catalog screens, dashboards, charts, and report views. React is appropriate for this type of interface because it supports declarative and component-based user interface development, allowing reusable interface parts to be built and maintained more efficiently (Meta Open Source, n.d.).

React.js is appropriate for RevInsight AI because the system requires dynamic dashboard updates and interactive analytics panels. Through reusable components, the developers can organize screens for revenue summaries, top-selling products, service labor rankings, and concentration results. JavaScript supports client-side interaction, validation, and data rendering, making the interface more responsive for users. Compared with static web pages, the use of React.js and JavaScript provides a more flexible frontend structure for analytics-based applications and supports the goal of presenting business information in a clear and usable format.

Backend Technologies

Backend technologies handle business logic, authentication, request processing, analytics computation, and communication between the interface and database. RevInsight AI uses Node.js and Express.js to provide server-side operations for transaction recording, catalog management, dashboard data retrieval, report preparation, and AI integration. Node.js is suitable because it is a JavaScript runtime used for server-side application development, while Express.js provides a lightweight web framework for building routes and APIs (Node.js contributors, n.d.; Express.js contributors, n.d.).

Node.js and Express.js support the proposed system by enabling structured communication between the frontend, database, analytics functions, and AI insight module. In RevInsight AI, the backend will process sales entries, service labor records, revenue summaries, and user requests from the frontend. This backend structure is appropriate for a web-based revenue monitoring system that requires consistent processing of daily business transactions and report generation, similar to the transaction and reporting needs discussed in web-based sales systems (Wahyudi & Hendriawan, 2024).

Database Technologies

Database technologies are necessary for storing, organizing, and retrieving business records. RevInsight AI uses MongoDB to store user accounts, product and accessory records, service categories, mechanic labor entries, sales transactions, analytics results, and generated reports. MongoDB stores records as document-based data, which is useful when transaction records may contain different combinations of item, service, labor, and report fields (MongoDB, n.d.).

MongoDB supports centralized storage of business data, allowing the system to retrieve records by date range, item category, service type, mechanic, and revenue source. This is important for calculating revenue distribution and concentration values. A centralized database also reduces the difficulty of searching through notebooks, receipts, or separate spreadsheets. For RevInsight AI, the database will serve as the foundation of accurate analytics, and the reliability of dashboards, reports, and OpenAI API-assisted insight summaries will depend on complete and consistent stored records.

Data Analytics and Visualization

Data analytics and visualization tools convert transaction records into meaningful business information. RevInsight AI uses analytics computation to calculate total revenue, category shares, top-performing products, top service labor, and concentration indicators. These outputs help the business owner determine which items and services contribute most to income.

Chart.js or a similar charting library is used to present revenue results through graphs, summary cards, rankings, and visual reports. Chart.js supports several chart types and responsive visualizations for modern web interfaces, making it suitable for presenting revenue summaries and concentration results (Chart.js contributors, n.d.). Dashboard literature and recent business intelligence studies also emphasize the value of performance dashboards and visual analytics in supporting decision-making (Eckerson, 2005; Gonçalves et al., 2023; Kobi, 2024). In RevInsight AI, analytics and visualization work together to transform raw sales records into dashboards and reports.

Artificial Intelligence Integration

Artificial intelligence integration in RevInsight AI is implemented through the OpenAI API, which will generate natural-language insights from summarized revenue analytics. The AI component does not replace the analytics engine and does not require the development of a new machine learning model. Instead, it explains computed results in a way that is easier for the business owner to understand. The OpenAI API supports text generation from prompts, which makes it suitable for turning computed revenue summaries into short business insight explanations (OpenAI, n.d.).

The AI feature receives summarized information such as top revenue sources, percentage contributions, sales trends, and concentration indicators. It then produces observations, decision-support notes, and possible areas for attention. AI-assisted decision support and visualization studies show that clear visual outputs and AI-supported explanations can help users interpret complex information for decision-making (Dachepalli, 2025; Neri et al., 2025).

For RevInsight AI, the AI integration supports practical decision-making by explaining whether revenue is concentrated in a few products or services. The generated output will be treated as decision support only, while the owner remains responsible for reviewing and applying the insights.

Reporting and Export Tools

Reporting and export tools support documentation and business review. RevInsight AI includes report generation features that allow users to produce summaries of revenue distribution, top-performing products and services, and AI-supported insights. These reports may be exported for recordkeeping, adviser validation, or client review.

The reporting function is important because business owners often need summarized information for planning, restocking, pricing, and service management.

Instead of manually compiling sales records, the system can generate structured reports based on stored transactions and computed analytics results.

Design and Development Tools

Design and development tools support planning, coding, testing, and collaboration. Figma is used to prepare wireframes and interface prototypes before system development. Visual Studio Code is used as the main code editor for frontend, backend, and integration work. GitHub is used for version control, source code backup, and team collaboration. These tools help the researchers organize the development process and maintain the system throughout the project. They also support documentation, revision, and progress tracking, which are important for adviser monitoring and system validation.

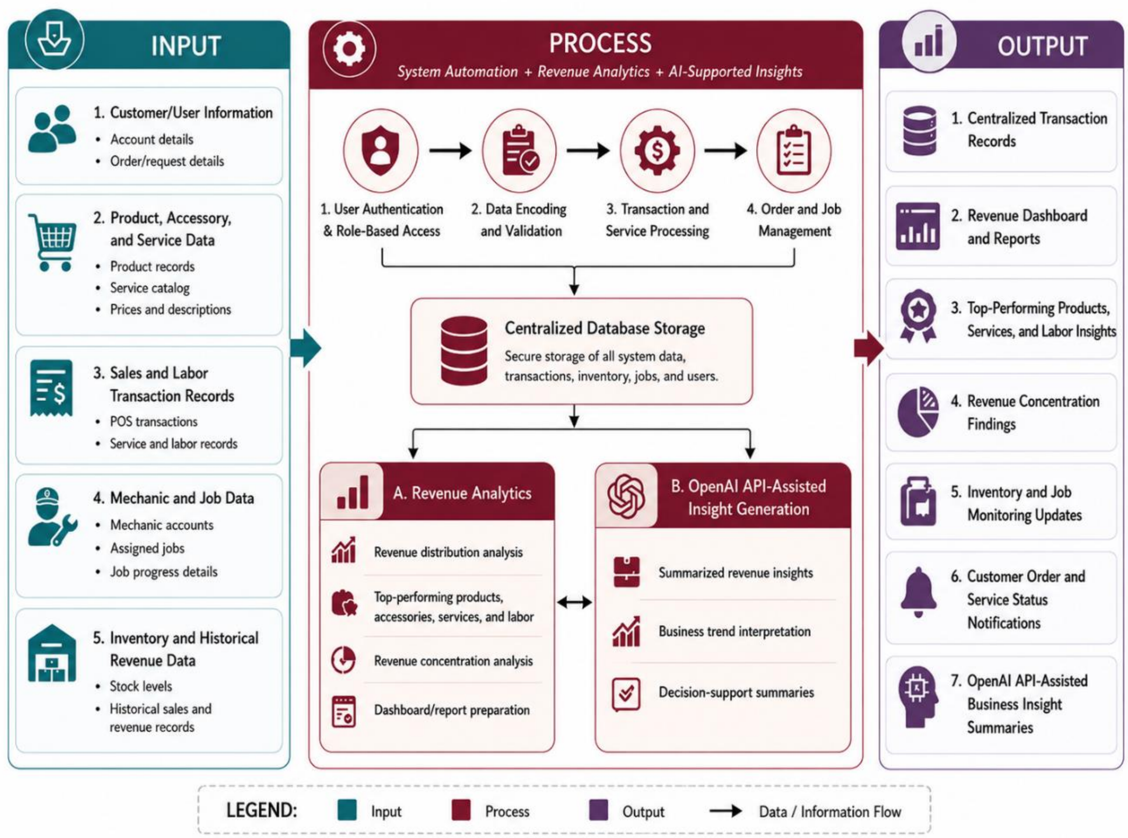
Deployment Technologies

Deployment technologies allow the system to be hosted and accessed through modern web browsers. Platforms such as Render, Hostinger, or another approved hosting service may be used to deploy the frontend and backend components of RevInsight AI. Deployment preparation ensures that the system can be demonstrated and evaluated in an environment similar to actual use. For the proposed system, deployment tools are important because the client needs access to dashboards, transaction forms, reports, and AI-supported/API-assisted insight summaries without relying on scattered local files. Web-based deployment supports accessibility, maintainability, and future improvement of the system.

2.3 Conceptual Framework

Figure 1

Conceptual Framework of RevInsight AI



The input stage includes sales transaction records, product and service details, historical revenue data, and user or business records because these are the primary data needed to evaluate how income is distributed across the business. Sales transaction records provide the basis for total revenue computation, while product and service details allow the system to classify income according to motorcycle parts, accessories, and mechanic labor services. Historical revenue data also helps compare revenue patterns across periods, making it possible to identify whether the business depends heavily on limited revenue sources.

In the process stage, the system validates, stores, organizes, and analyzes the collected data. The analytics module computes total revenue, revenue distribution, top-performing items and services, and concentration indicators. The AI-supported insight module receives summarized analytics results and generates natural-language observations that can help the owner understand revenue patterns and possible risks.

In the output stage, the system presents the processed information through dashboards, charts, reports, and AI-supported/API-assisted business insight summaries. These outputs allow the business owner to identify which products or services generate the highest revenue, monitor possible overdependence on limited income sources, and support decisions related to restocking, pricing, promotion, and service planning.

Overall, the conceptual framework shows how RevInsight AI transforms transaction records into useful business insights. The integration of analytics and AI-supported explanation improves the usefulness of the system while keeping the final decision-making responsibility with the business owner.

3. METHODOLOGY

This chapter presents the methods, procedures, and technical approach that will be used in the development and evaluation of RevInsight AI: AI-Driven Revenue Concentration Analysis System for Manoy's Motorcycle Parts, Accessories and Services. It discusses the research design, client requirements analysis, system architecture, AI integration design, development tools, implementation procedure, testing and evaluation procedures, evaluation methods, ethical considerations of the study.

3.1 Research Design

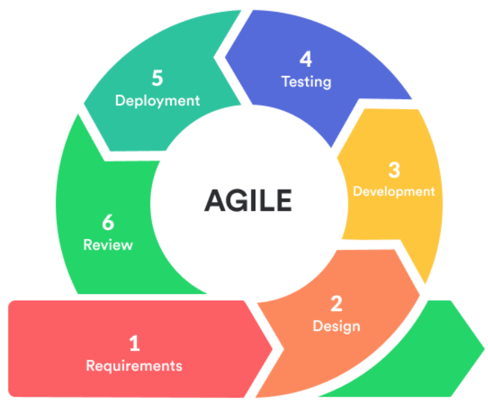
The study will use a developmental research design focused on the design, development, implementation, and evaluation of a web-based revenue concentration analysis system for Manoy's Motorcycle Parts, Accessories and Services. The project will follow an Agile development approach because it supports iterative development, regular inspection, adaptation, client feedback, and continuous improvement during software development (Schwaber & Sutherland, 2020).

The development process will include planning, requirements analysis, system design, development, AI integration, testing, deployment preparation, and evaluation. During planning and requirements analysis, the researchers will examine the current sales recording process of the business and identify the data needed to monitor revenue from motorcycle parts, accessories, and mechanic labor services. During the design phase, the researchers will prepare the system architecture, data storage structure, user interface, and required diagrams.

Figure 2 presents the Agile software development methodology that will guide the development of RevInsight AI.

Figure 2

Agile Development Model of the Study



As shown in Figure 2, the Agile process will involve requirements gathering, design, development, testing, deployment, and review. These stages will allow the researchers to work iteratively, gather feedback from the client, and improve system modules before final testing and evaluation.

The client will be involved throughout the development process by validating the business requirements, reviewing sample screens, checking whether the transaction fields reflect the actual sales and service workflow, and giving feedback on the dashboard outputs. The researchers will use the feedback to revise the forms, reports, analytics display, and AI-generated insight summaries. This iterative process will ensure that the system remains aligned with the actual operational needs of the business.

The final system will be evaluated through functional testing, usability testing, user acceptance testing, and performance testing. The evaluation will focus on whether the system can accurately record transactions, classify revenue sources, identify top-selling items and mechanic labor services, generate revenue concentration outputs, and provide useful AI-supported insights for business decision-making.

3.2 Client Requirements Analysis

Client requirements analysis will serve as the basis for the design and development of the proposed system. This section discusses the client profile, current system analysis, identified problems, functional requirements, non-functional requirements, and system context of RevInsight AI.

3.2.1 Client Profile

Manoy's Motorcycle Parts, Accessories and Services is a motorcycle-related business that sells motorcycle parts and accessories and provides service-related work such as repair, maintenance, installation, and mechanic labor. The business serves customers who purchase replacement parts, upgrade accessories, or request services from mechanics. Its operations involve daily transactions that may include item sales, labor fees, service charges, and combinations of product and service purchases.

The business currently depends on manual or semi-digital recording methods to monitor its sales transactions. While these methods can store basic transaction information, they do not provide a centralized and analytical way to track which items, accessories, or mechanic labor services generate the highest revenue. As a result, the owner has limited visibility over revenue concentration and must often rely on observation or manual checking to determine what sells best.

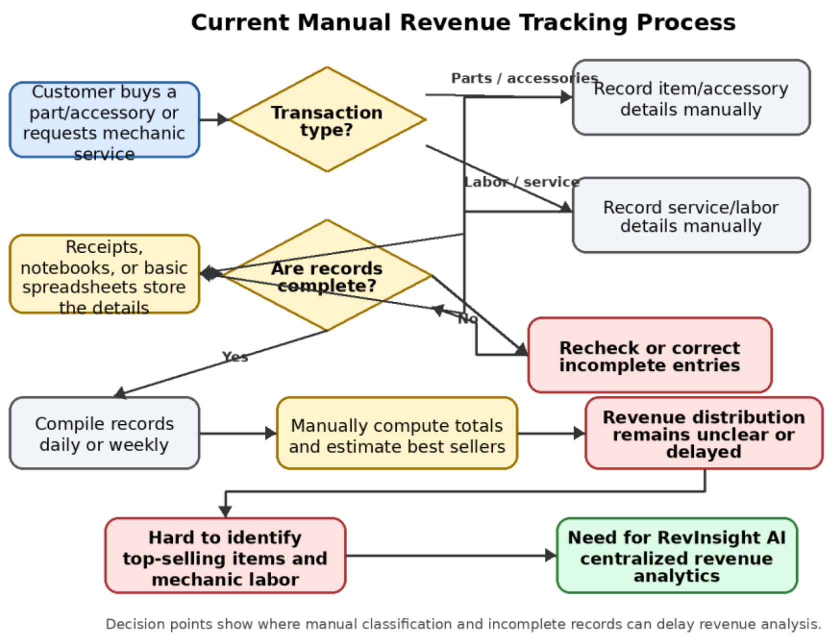
3.2.2 Current System Analysis

The current process begins when a customer buys motorcycle parts or accessories or requests mechanic labor. The transaction details are then recorded manually through notebooks, receipts, or basic spreadsheets. The owner or staff may later compile the records to compute total sales. However, the process does not automatically classify revenue according to item category, accessory type, mechanic labor, service type, or date range. Because of this, identifying the top-selling items and services requires manual review and may produce delayed or inaccurate results.

Figure 3 presents the current manual revenue tracking process of the business. It shows how sales and service transactions are recorded, checked for completeness, compiled, and reviewed manually before the owner can make business decisions.

Figure 3

Current Manual Revenue Tracking Process



As shown in Figure 3, the existing process relies heavily on manual recording and manual checking of revenue information. The decision points for transaction type and record completeness show where delays, incomplete entries, or duplicate checking may occur. This creates difficulty in monitoring revenue distribution, identifying best-selling items, determining which mechanic labor services are most profitable, and evaluating whether the business depends too much on limited revenue sources.

3.2.3 Problem Identification

The analysis of the existing manual or semi-digital process of Manoy's Motorcycle Parts, Accessories and Services revealed several issues. The business cannot easily track total revenue from motorcycle parts, accessories, and mechanic labor services in one centralized system. The owner also has difficulty identifying which items, accessories, or labor services are selling best within a specific period, while manual recording increases the possibility of incomplete entries, computation errors, duplicate records, and delayed sales summaries. Revenue sources are not clearly categorized, making it difficult to determine whether income is concentrated in only a few products or services. As a result, decisions on restocking, service planning, pricing, and mechanic workload management are often based mainly on estimation rather than real-time analytics.

3.2.4 Functional Requirements

The proposed RevInsight AI system will provide the following functionalities based on the identified system features:

- The system shall allow the Admin, Mechanic, and User/Customer to log in using authorized credentials.
- The system shall allow the Admin to manage users, roles, item, accessory, and service catalogs, mechanic accounts, and system records.
- The system shall allow the Admin to record sales and labor transactions through the POS module.
- The system shall allow the Admin to view dashboards, revenue reports, top-selling items, and revenue concentration analysis results.
- The system shall allow the Admin to generate OpenAI API-assisted business insight summaries and export records or reports.

• The system shall allow the Mechanic to view available jobs, current or assigned jobs, job details, and update job progress.

• The system shall allow the User/Customer to browse services, use POS-related ordering functions, and access customer-facing features.

Figure 4 shows the use case diagram of RevInsight AI. This diagram illustrates how the system interacts with its users, including the Admin, Mechanic, and User/Customer.

Figure 4

Use Case Diagram of RevInsight AI

As shown in Figure 4, the Admin can log in and perform management, POS, dashboard, transaction, revenue analytics, AI-supported insight, export, and backup functions. The Mechanic can access job-related functions such as available jobs, current or assigned jobs, service-related work, and dashboard access depending on authorized permissions. The User/Customer can access customer-facing functions such as services, POS-related ordering, and account-based access. The diagram shows the main functions of RevInsight AI and the role of each user in the system.

3.2.5 Non-Functional Requirements

The non-functional requirements define the quality attributes of RevInsight AI to ensure that it operates efficiently, securely, reliably, and in a user-friendly manner while providing a clear experience for its users.

Performance

- The system shall process login, transaction recording, dashboard loading, analytics generation, and report export within an acceptable response time during normal business use.

- The system shall support timely updates for transaction records, revenue summaries, and dashboard information.

Security

- The system shall require authorized login credentials for Admin and Mechanic users.

• The system shall use role-based access control to restrict features based on assigned user roles.

• The system shall protect business records, customer details, transaction data, and revenue information from unauthorized access.

Reliability

• The system shall maintain accurate recording and storage of product, service, labor, and transaction data.

• The system shall provide stable access to dashboards, reports, and stored records during operation.

Usability

• The system shall provide a clear and understandable interface for Admin, Mechanic, and User/Customer roles.

• The system shall allow users to navigate features such as POS, service browsing, job monitoring, dashboard viewing, and report generation with minimal difficulty.

Accuracy

• The system shall compute revenue totals, rankings, concentration values, and report summaries based on stored transaction data.

• The system shall minimize manual computation errors by using automated calculations and structured records.

Compatibility and Maintainability

- The system shall be accessible through modern web browsers on desktop and compatible mobile devices with internet access.

- The system shall be structured so that future developers can update features, categories, reports, or AI prompts when needed.

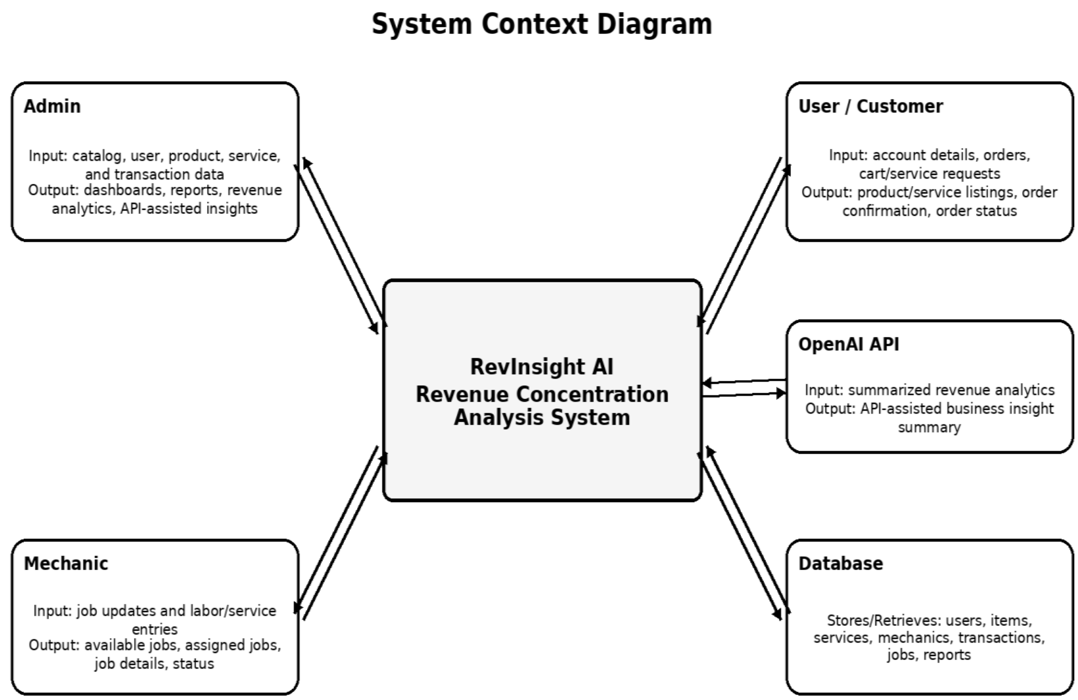
3.2.6 System Context

The system context defines the boundaries of RevInsight AI and shows how external entities interact with the proposed system. The main external entities are the Admin, Mechanic, User/Customer, centralized database, and OpenAI API service.

Figure 5 shows the context diagram of RevInsight AI. This diagram represents the system as a single process and shows its connection with the Admin, Mechanic, User/Customer, database, and OpenAI API service.

Figure 5

System Context Diagram of RevInsight AI



As shown in Figure 5, the Admin will provide catalog, user, product, service, and transaction data and will receive dashboards, reports, revenue analytics, and API-assisted insight summaries. The Mechanic will provide job updates and labor or service entries and will receive available jobs, assigned jobs, job details, and status information. The User/Customer will provide account details, orders, cart details, and service requests and will receive product or service listings, order confirmation, and order status. The system will store and retrieve validated records from the centralized database and will send summarized revenue analytics to the OpenAI API to generate API-assisted business insight summaries. Raw sensitive details will not be necessary for AI processing because aggregated values will be used whenever possible.

3.3 System Architecture

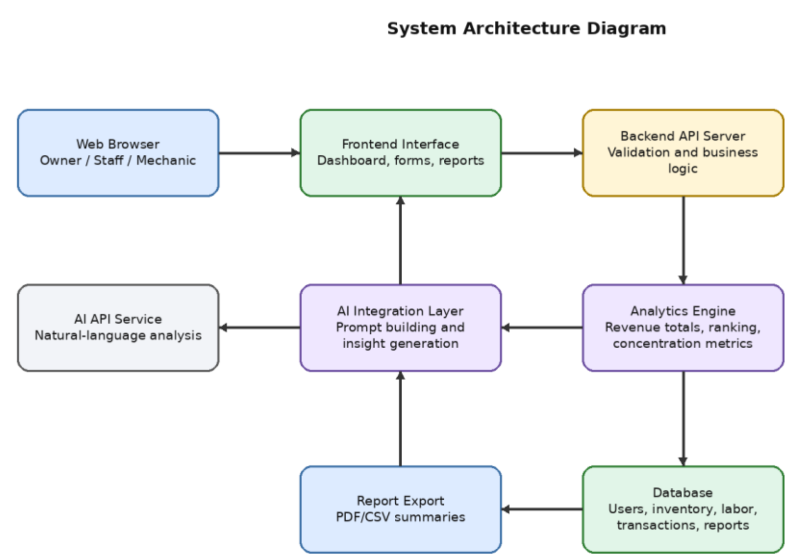
The proposed system will be designed as a web-based application composed of a user interface, backend server, database, analytics engine, AI integration layer, and report generation component. This structure will allow the business to store transactions centrally, process revenue data, generate visual analytics, and produce AI-supported explanations based on the recorded sales and labor information.

1. System Architecture Diagram

Figure 6 shows the system architecture of RevInsight AI. The architecture identifies the major components and their relationships within the proposed system.

Figure 6

System Architecture Diagram of RevInsight AI



As presented in Figure 6, users will access the system through a web browser. The frontend interface will display forms, dashboards, charts, and reports. The backend API server will handle validation, authentication, business logic, and data transfer. The database will store user accounts, item and service catalogs, transaction records, mechanic labor entries, and generated reports. The analytics engine will compute revenue totals, top-selling items, labor rankings, and concentration metrics, while the AI integration layer will generate text-based insights from summarized revenue results.

System Modules and Components

The system will include the following major modules and components:

- Authentication Module - manages login, user roles, account access, and session control.

- Catalog Management Module - manages motorcycle parts, accessories, service types, labor categories, prices, and mechanic information.

- Transaction Management Module - records product sales, accessory sales, service transactions, labor fees, quantities, dates, and assigned mechanics.

- Analytics Module - computes total revenue, percentage contribution, top-selling items, top labor services, revenue distribution, and concentration indicators.

- AI Insight Module - generates natural-language summaries, observations, and recommended focus areas based on analytics output.

- Dashboard and Reporting Module - displays charts, tables, summary cards, and exportable reports for owner review.

- Database Module - stores validated records and supports retrieval of historical revenue data for analysis.

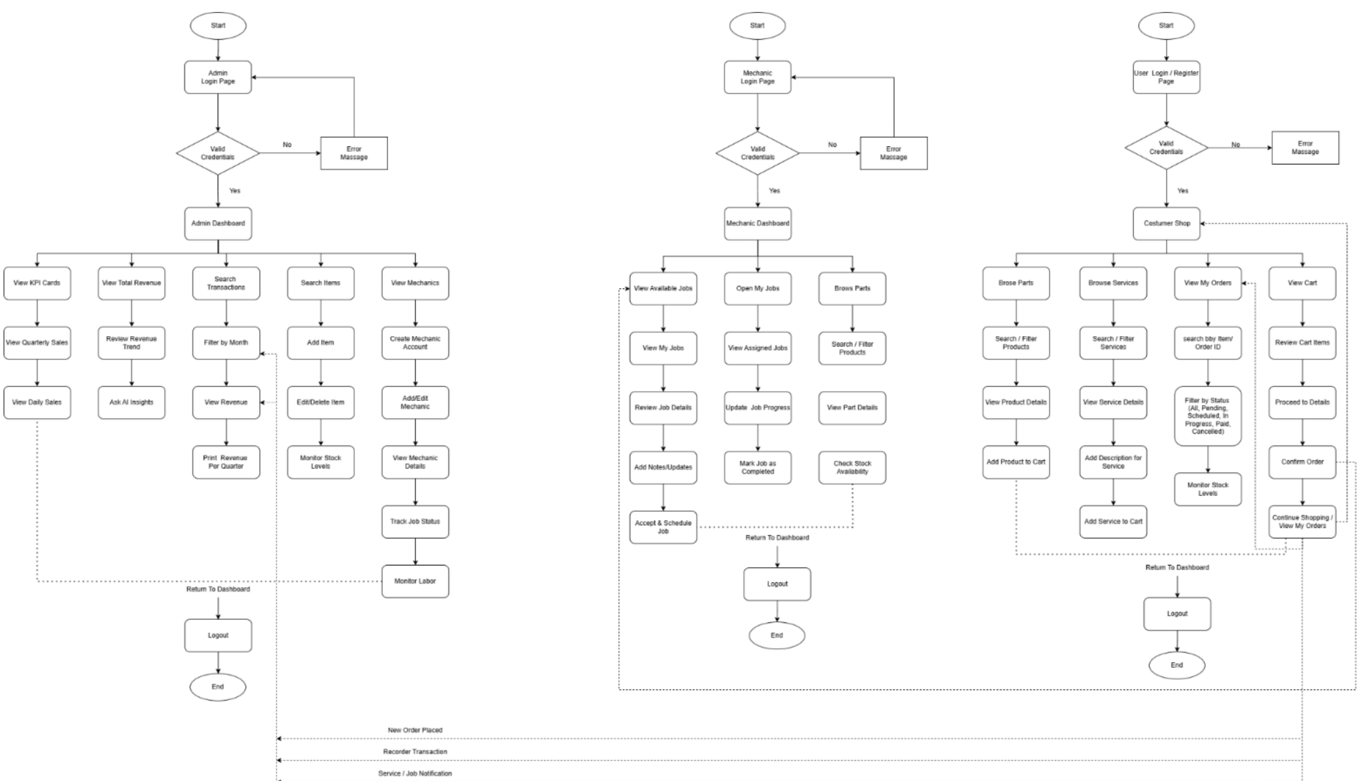
2. System Flow Diagram

The data flow of the system will begin with the encoding of sales and service transactions. After validation, the data will be stored in the database and processed by the analytics engine. Aggregated results will be displayed on the dashboard and may be sent to the AI integration layer for insight generation. Reports will then be generated for the owner or authorized users.

Figure 7 presents the system flow diagram of RevInsight AI. It shows how Admin, Mechanic, and User/Customer actions move through login validation, dashboard access, transaction and service processing, job handling, order management, revenue monitoring, and system logout.

Figure 7

System Flow Diagram of RevInsight AI



As shown in Figure 7, the system flow diagram presents the role-based workflow of RevInsight AI. The Admin can access dashboard, revenue monitoring, item management, transaction search, mechanic management, and AI insight functions. The Mechanic can access available jobs, assigned jobs, job details, job progress updates, and parts availability checking. The User/Customer can browse products and services, add items or services to the cart, place orders, view orders, and monitor order status. These workflows support the movement of transaction and service data into the system for revenue tracking, job monitoring, reporting, and decision-making.

User Roles

The proposed system will include defined user roles to control access and protect business data. The Admin will manage users, item and service catalogs, mechanic accounts, reports, dashboards, revenue analytics, system records, and AI-supported insight functions. The Mechanic will access available jobs, assigned jobs, job details, progress updates, and parts availability information based on authorized permissions. The User/Customer will access customer-facing features such as services, POS-related ordering, cart functions, order placement, and order status monitoring. The role structure will ensure that each user can access only the features needed for their responsibilities.

3.4 AI Integration Design

RevInsight AI will integrate the OpenAI API to generate natural-language insights from summarized revenue data. The AI component will not replace the analytics engine and will not train a new machine learning model. Instead, it will receive computed and aggregated results from the system, such as top-selling parts, top accessory categories, top service labor, revenue contribution percentages, and concentration indicators, then produce an easy-to-understand explanation for the business owner.

The purpose of the AI feature is to support decision-making by summarizing revenue patterns, identifying possible risks, and suggesting areas that may need attention. For example, if a large percentage of revenue comes from only a few items or a single labor service, the AI insight may explain that the business has a concentration risk and should monitor inventory availability or diversify service promotion. If mechanic labor revenue increases, the AI insight may suggest reviewing service capacity or mechanic scheduling.

The AI integration method will involve three steps. First, the analytics engine will compute and prepare summarized data from stored transactions. Second, the backend will format the summary into a controlled prompt that excludes unnecessary personal or sensitive information. Third, the OpenAI API will return a text-based insight that will be displayed on the dashboard or included in reports. Human review will remain necessary because the AI-generated output will be used only as a decision-support guide, not as a final business decision.

3.5 Development Tools and Technologies

The following tools and technologies will be used to develop RevInsight AI. These tools are selected to support web-based development, data storage, analytics processing, AI integration, and report generation.

Table 3. Development Tools and Technologies

Tool or Technology	Purpose in the System
HTML	Structures the web pages, forms, dashboard sections, and report layouts.
CSS / Tailwind CSS	Provides responsive styling and improves the visual layout of the user interface.
JavaScript	Supports client-side interactions, form validation, and dynamic dashboard behavior.
Node.js	Serves as the runtime environment for the backend application.
Express.js	Handles routes, API endpoints, request validation, and server-side logic.
MongoDB	Stores users, catalog records, transactions, service labor entries, and report data in a flexible database.
Charting Library	Displays revenue summaries, rankings, trends, and concentration charts on the dashboard.
OpenAI API	Generates AI-supported text insights from summarized revenue analytics.
Figma	Supports interface planning, wireframes, and prototype design.
Visual Studio Code	Serves as the main code editor for system development.
GitHub	Supports version control, source code backup, and collaboration.

3.6 Implementation Procedure

The implementation procedure describes the sequence of activities that will be followed in developing the system. The process will begin with requirements gathering and will end with testing, evaluation, and deployment preparation.

Table 4. Implementation and Procedures

Phase	Activities
Phase 1: Requirements Gathering	Interview the client, observe the current transaction recording process, identify revenue categories, and define the data needed for items, accessories, services, labor fees, and mechanics.
Phase 2: System Design	Prepare the system architecture, data storage structure, user interface layout, process flow, use case diagram, context diagram, and data flow diagram.
Phase 3: Data Storage Development	Create collections or tables for users, items, accessories, services, mechanics, transactions, analytics records, and reports.
Phase 4: Frontend Development	Develop login pages, transaction forms, catalog screens, dashboards, report views, and responsive interface components.
Phase 5: Backend Development	Develop authentication, transaction processing, validation, CRUD functions, analytics computation, and report generation endpoints.
Phase 6: AI Integration	Connect the backend to the OpenAI API, prepare summarized prompts, receive AI-generated insights, and display them in the dashboard or reports.
Phase 7: Testing and Revision	Conduct functional, usability, performance, and user acceptance testing; collect feedback; and revise the system based on results.
Phase 8: Deployment Preparation	Prepare system documentation, user guide, backup procedures, and final deployment requirements for client use.

3.7 Testing and Evaluation

Testing and evaluation will be conducted to determine whether RevInsight AI meets its objectives and performs correctly under expected use. The testing process will focus on the accuracy of transaction recording, correctness of revenue computation, usability of the interface, performance of dashboard loading, and usefulness of AI-generated insights.

The evaluators will be selected through purposive sampling because they must be directly related to the use or assessment of the proposed system. The target evaluators will include one (1) business owner or admin, two (2) mechanics or authorized staff, five (5) customer/users, and three (3) IT-related evaluators, for a total of eleven (11) respondents. They will be selected based on their familiarity with the business process, expected system use, or ability to assess system quality. Their feedback will be used to evaluate RevInsight AI in terms of functionality, usability, performance, accuracy, security, and overall usefulness.

Table 5. Testing and Evaluation Plan

Testing Method	**Purpose**	**Expected Output**
Functional Testing	To verify that login, catalog management, transaction recording, dashboard analytics, AI insights, and report generation function correctly.	All major system features will operate according to requirements.
Usability Testing	To determine whether the owner, staff, and mechanics can use the system easily and understand the screens and reports.	Users will rate the system as clear, understandable, and easy to use.
User Acceptance Testing	To confirm that the client accepts the system based on actual business needs and expected workflow.	The client will validate whether the system supports revenue tracking and decision-making.

Testing Method	Purpose	Expected Output
Performance Testing	To measure response time during transaction entry, dashboard loading, analytics generation, and report export.	The system will respond within acceptable time limits during normal use.
Accuracy Testing	To compare system-generated totals, rankings, and concentration results with manually computed sample data.	The system will produce correct computations based on recorded transactions.

The adviser will periodically review the progress of the project. The researchers will submit outputs such as diagrams, interfaces, data storage structure, module progress, testing documents, and revised versions for validation before proceeding to the next development phase.

3.8 Evaluation Methods

The evaluation results will be analyzed using descriptive analysis, a Likert scale, and weighted mean. These methods will help summarize user feedback on functionality, usability, reliability, performance, security, and overall acceptability of the system.

A five-point Likert scale will be used to interpret responses from evaluators. The scale will help determine the level of agreement or satisfaction of users with each evaluation statement.

Table 6. Five-Point Likert Scale Interpretation

Scale	Range	Verbal Interpretation
5	4.50 - 5.00	Strongly Agree / Excellent
4	3.50 - 4.49	Agree / Very Good

Scale	Range	Verbal Interpretation
3	2.50 - 3.49	Moderately Agree / Good
2	1.50 - 2.49	Disagree / Fair
1	1.00 - 1.49	Strongly Disagree / Poor

The weighted mean will be used to determine the average rating for each evaluation criterion. The formula will be: Weighted Mean = sum of weighted responses divided by the total number of responses. The computed results will be interpreted using the scale shown in Table 6. Descriptive analysis will then be used to explain the results and determine whether the system meets its objectives.

3.9 Ethical Considerations

The study will follow ethical practices in handling business data and system evaluation. The researchers will secure the consent of the client before collecting sample records, observing business processes, or conducting evaluation activities. The use of actual transaction data will be limited to what is necessary for system development and testing.

Data privacy will be observed by protecting business records, user accounts, prices, revenue summaries, and service-related information. Access to the system will be limited to authorized users. If sample data are needed for demonstration or testing, the researchers may use anonymized or dummy records to avoid exposing confidential business information.

Responsible use of AI will also be observed. The AI feature will be used only to generate decision-support insights from summarized and non-sensitive revenue analytics.

It will not be used to make automatic business decisions, evaluate employee performance unfairly, or expose private customer or business information. The owner will remain responsible for reviewing and applying the generated insights.

REFERENCES

Alsulami, A., Hamid, S., & Ghani, N. (2024, December). *Organizational Factors Influencing Data Analytics Adoption in SMEs: A Systematic Review*. igi-global. https://www.igi-global.com/gateway/article/full-text-html/356673

Ardito, L., Filieri, R., Raguseo, E., & Vitari, C. (2024, November 24). *Artificial intelligence adoption and revenue growth in European SMEs: synergies with IoT and big data analytics*. ScienceDirect. https://www.sciencedirect.com/org/science/article/pii/S1066224324000121#ref031

Bintang, M. K. (2026, March 13). *Design and Development of an integrated ordering and sales analysis system at Kaluna living*. Journal Lemlit. https://jurnal.ubhinus.ac.id/index.php/SMATIKA/article/view/2179/1145

Campbell, J., & Oyinloye, P. (2024, November 15). *Leveraging data analytics for financial stability: A blueprint for sustaining SMEs in economically distressed regions of the U.S*. CARI Journals. https://carijournals.org/journals/index.php/IJPID/article/view/2288

Chart.js. (n.d.). Open source HTML5 Charts for your website. https://www.chartjs.org/docs/latest/

Dachepalli, V. (2025, March 25). *AI-Driven Decision Support Systems in ERP*. Sciforce. https://doi.org/10.55124/csdb.v2i2.248

Drydakis, N. (2022, March 4). *Artificial intelligence and reduced SMEs' business risks. A dynamic capabilities analysis during the COVID-19 pandemic*. SpringerLink. https://link.springer.com/article/10.1007/s10796-022-10249-6

Elumilade, O. O., Ogundeji, I., Achumie, G., & Omokhoa, H. (2025, March 25). *Leveraging financial data analytics for business growth, fraud prevention, and risk mitigation in markets*. FE Gulf Publishers. https://fegulf.com/index.php/gjabr/article/view/119

Express. (n.d.). Node.js web application framework. https://expressjs.com/

General Data Privacy Principles. (n.d.). National Privacy CommissionNational Privacy Commission. https://privacy.gov.ph/data-privacy-act/#w11

Gonçalves, C., Gonçalves, M., & Campante, M. (2023, November 15). *Developing integrated performance dashboards visualisations using power BI as a platform*. MDPI. https://www.mdpi.com/2078-2489/14/11/614

Kobi, J. (2026, May 31). *Developing dashboard analytics and visualization tools for effective performance management and continuous process improvement| International Journal of innovative science and research technology*. International Journal of Innovative Science and Research Technology. https://www.ijisrt.com/developing-dashboard-analytics-and-visualization-tools-for-effective-performance-management-and-continuous-process-improvement

Neri, G., Marshall, S., Chan H, H., Yaghi, A., Tabor, D., Sinha, R., & Mazumdar, S. (2025, August 14). *Data visualization in AI-assisted decision-making: A systematic review*. Frontiers. https://www.frontiersin.org/journals/communication/articles/10.3389/fcomm.2025.1605655/full

Node.js v26.1.0. (n.d.). Node.js Run JavaScript Everywhere. https://nodejs.org/api/documentation.html

Nurdin, A., Salmi, G., Sentosa, K., Wijayanti, A., & Prasetya, A. (2023, January). *Utilization of business intelligence in sales information systems | Journal of information system exploration and research*. SHM Publisher. https://shmpublisher.com/index.php/joiser/article/view/101

OARJMS, E. (2024, September 30). *Forecasting financial stability in SMEs: A comprehensive analysis of strategic budgeting and revenue management*. Open Access Research Journal of Multidisciplinary Studies. https://oarjpublication.com/journals/oarjms/content/forecasting-financial-stability-smes-comprehensive-analysis-strategic-budgeting-and-revenue

Ogbuefi, E., Mgbame, A., Akpe, O., Abayomi, A., & Adeyelu, O. (2024, June). *Operationalizing SME growth through real-time data visualization and analytics*. International Journal of Advanced Multidisciplinary Research and Studies. https://www.multiresearchjournal.com/arclist/list-2024.4.6/id-4251?utm_source=copilot.com

Oldemeyer, L., Jede, A., & Teuteberg, F. (2024, February 1). *Investigation of artificial intelligence in SMEs: A systematic review of the state of the art and the main implementation challenges*. SpringerLink. https://link.springer.com/article/10.1007/s11301-024-00405-4

OpenAI API. (n.d.). OpenAI Developers. https://developers.openai.com/api/docs/guides/text

Panigrahi, R., Shrivastava, A. K., & Nudurupati, S. (2024, March 29). *Impact of inventory management on SME performance: a systematic review*. Emerald Insight. https://doi.org/10.1108/IJPPM-08-2023-0428

Perdana, A., Lee, H., Koh, S., & Arisandi, D. (2022, March). *Data analytics in small and mid-size enterprises: Enablers and inhibitors for business value and firm performance*. ScienceDirect. https://www.sciencedirect.com/science/article/abs/pii/S146708952100049X?via%3Dihub

React. (n.d.). Meta Open Source. https://opensource.fb.com/projects/react/

Shao, T., Ngobeni, M., Mathabela, L., & Thango, B. (2025, May 7). *Applications and competitive advantages of data mining and business intelligence in SMEs performance: A systematic review*. MDPI. https://www.mdpi.com/2673-7116/5/2/22

Wahyudi, H. (2023, December 30). *Web based sales transaction and report system design: Study at a motorcycle spare parts company*. Journal Lemlit. https://journal.lemlit.org/index.php/ebmtj/article/view/82

Zamil, M. (2025, June 10). *AI-driven business analytics for financial forecasting: A systematic review of decision support models in smes*. Review of Applied Science and Technology. https://rast-journal.org/index.php/RAST/article/view/8